

Lecture Notes I

Course: General Topology

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Topic: Topological Spaces

Proof 2.7 (Characterisation of open and closed subsets).

i) If $p \in \text{Int} A$, then by definition $p \in U$ for some open $U \subseteq X$ with the property that $U \subseteq A$ and thus U is a neighbourhood of p contained in A by definition. Conversely, if p has a neighbourhood U contained in A , then U is open, $p \in U$ and $U \subseteq A$, so that clearly, $p \in \text{Int} A$.

ii) Write

$$\mathcal{B} = \{B \subseteq X : B \supseteq A \text{ and } B \text{ is closed in } X\},$$

then by definition

$$\overline{A} = \bigcap_{B \in \mathcal{B}} B,$$

and hence the complement is

$$\text{Ext} A = X \setminus \overline{A} = \bigcup_{B \in \mathcal{B}} (X \setminus B).$$

This way, it is clear that if $p \in \text{Ext} A$, then $p \in X \setminus B$ for some $B \in \mathcal{B}$, i.e., some B that is closed (which makes $X \setminus B$ open) and $B \supseteq A$, this implies that $X \setminus B \subseteq X \setminus A$ and hence $X \setminus B$ is an open neighbourhood contained in $X \setminus A$. Conversely, if p has an open neighbourhood V contained in $X \setminus A$, then $X \setminus V$ is closed and $X \setminus V \supseteq A$ so that $X \setminus V \in \mathcal{B}$ and thus $p \in \text{Ext} A$.

iii) We have that $p \in \partial A$ is equivalent to state that $p \notin \text{Int} A$ and $p \notin \text{Ext} A$, i.e., every neighbourhood of p is not contained in A and every neighbourhood of p is not contained in $X \setminus A$, respectively, these conclusions imply that for every neighbourhood V of p there exists a point ν such that $\nu \in V$ and $\nu \notin A$ (from $p \notin \text{Int} A$) and there exists a point η such that $\eta \in V$ and $\eta \notin X \setminus A \iff \eta \in A$ (from $p \notin \text{Ext} A$). We therefore conclude that every neighbourhood of p contains a point in A and a point in $X \setminus A$.

iv) The statement is equivalent to saying that a point is not in $\overline{A}$ if and only if it has a neighbourhood that does not contain a point in A . This is what we prove. By definition, a point is not in $\overline{A}$ if and only if it is in $\text{Ext} A$ so that we use ii) to see that indeed, a point p is not in $\overline{A}$ if and only if it has a neighbourhood contained in $X \setminus A$, i.e. the neighbourhood does not contain points in A .

v) We begin by showing that if $p \notin A$ and $p \notin \partial A$, then $p \notin \overline{A}$, i.e., $\overline{A} \subseteq A \cup \partial A$. Suppose $p \notin \partial A$ and $p \notin A$, then by iii) we have that there exists a neighbourhood V of p such that it either contains no points of A or contains no points in $X \setminus A$, but given that $p \notin A$ and $p \in V$ (the latter is by definition of neighbourhood), we conclude that V contains at least one point (p) of $X \setminus A$ and thus V contains no points in A , meaning that $p \notin \overline{A}$ by iv).

Conversely, if $p \notin \overline{A}$, then we seek to show that $p \notin A$ and $p \notin \partial A$, so assume that $p \notin \overline{A}$, then by iv) we have that there exists a neighbourhood of p that does not contain points in A , then by iii), we automatically have $p \notin \partial A$. Next, if $p \in A$, then every neighbourhood V of p automatically contains p , i.e., a point in A so that $p \in \overline{A}$, i.e., $p \in A \implies p \in \overline{A}$ and equivalently, $p \notin \overline{A} \implies p \notin A$. We therefore conclude that $p \notin \overline{A}$ implies $p \notin \partial A$ and $p \notin A$. These proofs show that $\overline{A} = A \cup \partial A$.

For the other equality, notice that by our previous remarks we have $\text{Int} A \subseteq A$ so that $\text{Int} A \cup \partial A \subseteq A \cup \partial A = \overline{A}$ and thus the first implication is immediate. For the reverse contention, suppose that $p \notin \text{Int} A$ and $p \notin \partial A$, then for any neighbourhood V of p there exists a point γ such that we have $\gamma \in V$ and $\gamma \notin A$, this means that there does not exist a neighbourhood of p with no points in $X \setminus A$. The assumption $p \notin \partial A$ is equivalent to state that there exists a neighbourhood of p that either has no points in A or it has no points in $X \setminus A$, but by the previous conclusion we

see that there does not exist a neighbourhood with the second property, and thus we conclude that there exists a neighbourhood of p that contains no points in A , i.e., $p \notin \bar{A}$. This shows that $\bar{A} \subseteq \text{Int}A \cup \partial A$ and the proof is done.

- vi) $\text{Int}A$ is open because it is the union of opens. Similarly, $\bar{A}$ is closed because it is the intersection of closed sets. It is immediate that $\text{Ext}A$ is open, and since we can write $\partial A = (X \setminus \text{Int}A) \cap (X \setminus \text{Ext}A)$, it is also immediate that ∂A is closed.
- vii) 1. $d) \iff b)$ This is just i).
2. $b) \Rightarrow c)$ Suppose that $A = \text{Int}A$, then for A to contain at least one of its boundary point we would require $b \in A$ and $b \in \partial A$, but $b \in \partial A$ is equivalent to $b \in X \setminus \text{Int}A$ and $b \in X \setminus \text{Ext}A$, and then, b cannot be in $\text{Int}A = A$, i.e., b is not in A so A contains none of its boundary points.
3. $c) \Rightarrow b)$ The statement A contains none of its boundary points is equivalent to say that $g \in A$ implies $g \in X \setminus \partial A$, that is, $g \in \text{Int}A$ or $g \in \text{Ext}A$, but given that $g \in A$, any neighbourhood V of g contains at least g itself so that it is not true that g has a neighbourhood contained in $X \setminus A$, i.e. g is not in $\text{Ext}A$. We conclude that $g \in \text{Int}A$ and hence $A \subseteq \text{Int}A$. Conversely, if $g \in \text{Int}A$, then by i), there is some neighbourhood V of g with $V \subseteq A$, but by definition $g \in V$, it is thus immediate that $g \in A$ and hence $\text{Int}A \subseteq A$. We therefore conclude that $A = \text{Int}A$.
4. $b) \Rightarrow a)$ This is just the first statement in vi).
5. $a) \Rightarrow b)$ If A is open, then A is an open neighbourhood of each of its points, so that $p \in A$ automatically implies $p \in \text{Int}A$, i.e. $A \subseteq \text{Int}A$, and $\text{Int}A \subseteq A$ is always true as we saw in the last part of point 3.
- viii) The proof is done in a similar fashion as in vii).

Proof 2.8. Problem Sheet

Proof 2.9 (Properties of continuous functions). Let X, Y, Z be topological spaces. Then

- i) If $f : X \rightarrow Y$ is a constant map, then there exists $c \in Y$ such that $f(x) = c$ for all $x \in X$. Let U_y be open in Y and then

$$f^{-1}(U_y) = \begin{cases} X & c \in U_y, \\ \emptyset & c \notin U_y, \end{cases}$$

so that $f^{-1}(U_y)$ is always open and therefore f is continuous.

- ii) For $\text{Id}_X : X \rightarrow X$ we have that $\text{Id}_X^{-1}(U) = U$ for any U , in particular, if U is open, then $\text{Id}_X^{-1}(U)$ is also open and hence Id_X is continuous.
- iii) Let $f : X \rightarrow Y$ be continuous, and let $f|_U$ be its restriction with U open in X . Then it is easy to see that for any $V \subseteq Y$

$$f|_U^{-1}(V) = U \cap f^{-1}(V),$$

and in particular, if V is open in Y , then continuity of f implies that $f^{-1}(V)$ is open and since topologies are closed under finite intersections, we conclude that $f|_U^{-1}(V)$ is open and thus $f|_U$ is continuous.

- iv) The conclusion is immediate from the fact that $(g \circ f)^{-1}(U) = f^{-1}(g^{-1}(U))$.

Proof 2.10 (Continuity is a local property). If f is continuous, then for any $x \in X$ take X to be the open neighbourhood of x for which it is readily seen that $f|_X$ is continuous. Conversely, suppose that for any $x \in X$ there exists a neighbourhood V of x on which the restriction $f|_V$ of f is continuous. Let U be open in Y and take any $x \in f^{-1}(U)$, then let V be the neighbourhood on which f is continuous, then $f|_V^{-1}(U)$ is open, but notice that since $x \in V$, then $f(x) = f|_V(x)$ for such x and moreover, the assumption $x \in f^{-1}(U)$ implies that $f(x) \in U$, i.e. $f|_V(x) \in U$ and therefore $x \in f|_V^{-1}(U)$, and also

$f|_V^{-1}(U) = V \cap f^{-1}(U) \subseteq f^{-1}(U)$. We therefore conclude that $f|_V^{-1}(U)$ is an open neighbourhood of x contained in $f^{-1}(U)$ and since this holds for any $x \in f^{-1}(U)$, we conclude that $f^{-1}(U)$ is open and the proof is done.

Proof 2.12 (Homeomorphisms are open). Notice that f has an inverse, call it g . Then for any $U \subseteq X$ we have

$$g^{-1}(U) = f(U)$$

so that $g^{-1}(\mathcal{T}_X) = f(\mathcal{T}_X)$ and therefore

$$f(\mathcal{T}_X) \subseteq \mathcal{T}_Y \iff g^{-1}(\mathcal{T}_X) \subseteq \mathcal{T}_Y.$$

The first inclusion means that f is open and the second one means that g is continuous.

Proof 2.13 (Properties of local homeomorphisms). a) If f is a homeomorphism, then for any $x \in X$ take X to be an open neighbourhood of x , so that $f|_X = f$ and $f(X) = Y$ because f is surjective, then it is clear that $f|_X : X \rightarrow f(X)$ is a homeomorphism.

b) Given any local homeomorphism f , the local criterion for continuity implies that f is continuous, and it is open since given any open V , we

c) If f is a bijective local homeomorphism, then it is continuous and open by b) and thus f is a homeomorphism (a continuous bijective map is a homeomorphism if and only if it is open).

Proof 2.14 (Sequences are not well-behaved in arbitrary topological spaces). Let Y be an indiscrete space, then take any sequence (y_n) in Y , given any $y \in Y$ we have that the only open neighbourhood of y is Y itself, so given “any” open neighbourhood U of Y , choose $N = 1$ so that $y_n \in U (= Y)$ for $n \geq 1$, this implies that every sequence in Y converges to each point in Y .

Proof 2.15 (Every metric space is Hausdorff). Given any metric space, take $p \neq q$, which by definition of metric implies that $d(p, q) > 0$ and thus $B_\delta(p), B_\delta(q)$ with $\delta = d(p, q)/2$ are the required open neighbourhoods of p, q , respectively.

Proof 2.16 (Properties of Hausdorff spaces). i) Let $q \in X$, then we show that $\{q\}$ is closed. Since X is Hausdorff, for any $p \in X \setminus \{q\}$ ($p \neq q$) there exist open neighbourhoods U_p, U_q of p, q with $U_p \cap U_q = \emptyset$, i.e., $U_p \subseteq X \setminus \{q\}$ so that every point of $X \setminus \{q\}$ has an open neighbourhood contained in it, that is, $X \setminus \{q\}$ closed.

ii) Take any sequence (x_n) in X which converges to x . Suppose that $x \neq x'$ and that $(x_n) \rightarrow x'$, then choose open neighbourhoods U, U' of x, x' , respectively, such that $U \cap U' = \emptyset$. Definition of convergence implies that there exist K, K' such that $x_n \in U$ and $x_n \in U'$ for $n \geq K$ and $n \geq K'$, respectively, and thus $x_{K+K'} \in U \cap U'$, a contradiction. We therefore conclude that either $x = x'$ or (x_n) does not converge to x' , which means that the limit is unique (in case of existing).

Proof 2.17 (Basis criterion for an open subset). Let $\mathcal{B}$ be a basis for X . If a U is open, then any point $p \in U$ has an open neighbourhood contained in U meaning that $V = \bigcup \mathcal{G}$ for some $\mathcal{G} \subseteq \mathcal{B}$ (where V is the open neighbourhood of p for which $V \subseteq U$), then clearly $p \in G$ for some $G \in \mathcal{G} \subseteq \mathcal{B}$ and hence $p \in G \subseteq V \subseteq U$. Conversely, if for any given $p \in U$ there is some $B \in \mathcal{B}$ such that $p \in B \subseteq U$, then this means that any point in U has an open neighbourhood (B) contained in U , which is the definition of U being open.

Proof 2.19 (Continuity is fully characterised by a basis). Let $\mathcal{B}$ be a basis for Y . If f is continuous, then the preimage of any open is open, so that the conclusion is immediate from the definition of basis. Conversely, suppose that $f^{-1}(B)$ is open in X for each $B \in \mathcal{B}$, then given any open $U \subseteq Y$, we take $x \in f^{-1}(U)$, then we’ve showed that U satisfies the basis criterion with respect to $\mathcal{B}$, meaning that for $f(x) \in U$ there is a basis set $B \in \mathcal{B}$ such that $B \subseteq U$. This implies that $f(x) \in B$ and thus any point of $f^{-1}(U)$ has a neighbourhood contained in $f^{-1}(U)$, i.e., $f^{-1}(U)$ is open.

Proof 2.20 (Equivalent conditions for being the basis of some topology). ($\Rightarrow$) Given any X , suppose that $\mathcal{B}$ is a basis for some topology, then by definition of basis, we have that X (because it is open) is the union of some collection of elements in $\mathcal{B}$, this is property i). For any $B_1, B_2 \in \mathcal{B}$, which are open by definition of basis, we have that $B_1 \cap B_2$ is also open, which is equivalent to claim that for

each $x \in B_1 \cap B_2$ there exists an open neighbourhood, call it B_3 , of x contained in $B_1 \cap B_2$, this is property ii).

($\Leftarrow$) Suppose $\mathcal{B}$ is a collection of subsets satisfying

i) $\bigcup \mathcal{B} = X$.

ii) If $B_1, B_2 \in \mathcal{B}$ and $x \in B_1 \cap B_2$, there exists an element $B_3 \in \mathcal{B}$ such that $x \in B_3 \subseteq B_1 \cap B_2$.

Define the collection $\mathcal{T}$ as

$$\mathcal{T} = \left\{ \bigcup \mathcal{C} : \mathcal{C} \subseteq \mathcal{B} \right\},$$

then it is clear that $X, \emptyset \in \mathcal{T}$ (the first because of i) and the second being the empty collection). If $\{G_\alpha\}_{\alpha \in I}$ is any collection of elements in $\mathcal{T}$, then for each $\alpha \in I$ we have

$$G_\alpha = \bigcup \mathcal{C}_\alpha, \quad \mathcal{C}_\alpha \subseteq \mathcal{B},$$

which on declaring $\mathcal{C} = \bigcup_{\alpha \in I} \mathcal{C}_\alpha$ (this collection is such that $\mathcal{C} \subseteq \mathcal{B}$) we get

$$\bigcup_{\alpha \in I} G_\alpha = \bigcup \mathcal{C},$$

and hence $\bigcup_{\alpha \in I} G_\alpha \in \mathcal{T}$. Next, take any two $U_1, U_2 \in \mathcal{T}$, then $U_1 = \bigcup \mathcal{C}_1, U_2 = \bigcup \mathcal{C}_2$ ($\mathcal{C}_1, \mathcal{C}_2 \subseteq \mathcal{B}$) by definition, and for any given $x \in U_1 \cap U_2$ we have $x \in U_1$ and $x \in U_2$, so that $x \in B_{x,1}$ and $x \in B_{x,2}$ for some $B_{x,1} \in \mathcal{C}_1, B_{x,2} \in \mathcal{C}_2$, but since $\mathcal{C}_1, \mathcal{C}_2 \subseteq \mathcal{B}$, we have that $B_{x,1}, B_{x,2} \in \mathcal{B}$ and by ii) we conclude that there exists $B_{x,3} \subseteq B_{x,1} \cap B_{x,2}$ such that $x \in B_{x,3}$. For any $x \in U_1 \cap U_2$, let $B_{x,3}$ be as in this construction. We claim that

$$U_1 \cap U_2 = \bigcup_{x \in U_1 \cap U_2} B_{x,3}.$$

For $U_1 \cap U_2 \subseteq \bigcup_{x \in U_1 \cap U_2} B_{x,3}$ take any $u \in U_1 \cap U_2$, then by the previous remark we have $u \in B_{u,3}$ for some $B_{u,3}$ as in the previous construction, i.e., $u \in \bigcup_{x \in U_1 \cap U_2} B_{x,3}$. Conversely, for $U_1 \cap U_2 \supseteq \bigcup_{x \in U_1 \cap U_2} B_{x,3}$ take any $u \in \bigcup_{x \in U_1 \cap U_2} B_{x,3}$, then $u \in B_{x,3}$ for some $B_{x,3}$ with $x \in B_{x,3} \subseteq B_{x,1} \cap B_{x,2} \subseteq U_1 \cap U_2$, i.e., $u \in U_1 \cap U_2$. We therefore conclude that $U_1 \cap U_2$ is the union of some collection of elements in $\mathcal{B}$ and thus $\mathcal{T}$ is a topology on X . By definition of basis for a topology, it is clear that $\mathcal{T}$ is the only topology for which $\mathcal{B}$ is a basis.

Proof 2.22 (Properties of Second Countable Spaces). Let X is a second countable space and let $\mathcal{B}$ be a countable basis of X .

i) Given any $p \in X$ take any neighbourhood V of p , then $V = \bigcup \mathcal{A}$ for some $\mathcal{A}$, in particular, given that $p \in V$, we have that $p \in A_p$ for some $A_p \in \mathcal{A}$ so let

$$\mathcal{B}_p = \{A \in \mathcal{A} : p \in A\} \neq \emptyset$$

then $\mathcal{B}_p \subseteq \mathcal{A} \subseteq \mathcal{B}$ so $\mathcal{B}$ is countable and is contained in V by definition, we therefore conclude that $\mathcal{B}_p$ is a countable neighbourhood basis for X at p and since p was arbitrary, X is first countable.

ii) TBD

iii) Let $\mathcal{U}$ be any open cover of X and define

$$\mathcal{B}' = \{B \in \mathcal{B} : \exists U \in \mathcal{U}, B \subseteq U\}$$

which is immediately countable due to $\mathcal{B}' \subseteq \mathcal{B}$. Now, for each $B \in \mathcal{B}'$ choose U_B to be the set for which $B \subseteq U_B \in \mathcal{U}$, then the collection

$$\mathcal{U}' = \{U_B : B \in \mathcal{B}'\}$$

is clearly a subset of $\mathcal{U}$ and thus countable. We now show that $\mathcal{U}'$ is still an open cover for X : For any given $p \in X$, we know that $p \in U_0$ for some $U_0 \in \mathcal{U}$ because $\mathcal{U}$ covers X , then since

$\mathcal{B}$ is a basis, p satisfies the basis criterion with respect to $\mathcal{B}$ so that there exists $B \in \mathcal{B}$ with $p \in B \subseteq U_0$ so $B \in \mathcal{B}'$ ($U_0 \in \mathcal{U}$) so that there is some $U_B \in \mathcal{U}'$ such that $p \in B \subseteq U_B$ (at least U_0 is one such set), this shows that $\mathcal{U}'$ is a countable subcover of $\mathcal{U}$ because p was arbitrary.

Proof 2.24 (Equivalences for being locally Euclidean). It is clear that any space for which a) or b) holds is locally Euclidean of dimension n ($\mathbb{R}^n$ and open balls in $\mathbb{R}^n$ are both open). Note that any open ball is homeomorphic to $\mathbb{R}^n$ itself, so we only need to show a). Conversely, suppose that M is locally Euclidean of dimension n , then given any $p \in M$ let U be the open neighbourhood that admits an element $\varphi \in \text{Homeo}(U, V)$ for some V open in $\mathbb{R}^n$, but since V is open in $\mathbb{R}^n$, let B be some ball centred at $\varphi(p)$ and contained in V , then by definition $\varphi(p) \in B$ and thus $p \in \varphi^{-1}(B)$, but $\varphi^{-1}(B)$ is open since φ is a homeomorphism and B is open, so $\varphi^{-1}(B)$ is an open neighbourhood of p homeomorphic to its image $\varphi(\varphi^{-1}(B)) = B$.

Proof 2.25 (Open subsets of manifolds are manifolds). Let M be a manifold and let V an open neighbourhood of M , then given any $p \in M$ there is a neighbourhood U of p homeomorphic to some open subset of $\mathbb{R}^n$ so $U \cap V$ is still open in M and still homeomorphic to some open subset and clearly, contained in V meaning that V is locally Euclidean (when considered as a topological space on its own) and any open subset of a Hausdorff [respectively, second countable] space is Hausdorff [respectively, second countable].